

A dispersion-corrected DFT study on adsorption of battery active materials anthraquinone and its derivatives on monolayer graphene and h-BN†

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Cite this: *J. Mater. Chem. A*, 2014, 2, 8910

9,10-Anthraquinone (AQ) and its derivatives, *i.e.*, benzofuro[5,6-*b*]furan-4,8-dione (BFFD), benzo[1,2-*b*:4,5-*b'*]dithiophene-4,8-dione (BDTD) and pyrido[3,4-*g*]isoquinoline-5,10-dione (PID), are environmentally friendly and cheap electrode materials. However, their significant solubility in electrolyte solutions limits the cycle performance of lithium-ion batteries. In this work a comparative investigation of these four organic molecules adsorbed on monolayer graphene and hexagonal boron nitride (h-BN) has been carried out using van der Waals (vdW) dispersion-corrected density-functional theory (DFT). The calculated results indicate that the vdW dispersion contributes to more than 80% of the total attractive interaction for all the complexes studied. The binding energies range from 1.06 to 1.31 eV, showing strong physisorption. The calculated binding energies of the four organic molecules are in the order: BFFD < BDTD < AQ < PID on monolayer graphene and BFFD < BDTD < PID < AQ on monolayer h-BN. The physisorption causes a work function shift relative to the isolated graphene or h-BN nanosheet in the order: AQ < BDTD < BFFD < PID on both the graphene and h-BN nanosheets. This sequence is dominated by the work functions of the four organic molecules. The strong physisorption suggests that the solubility of the four organic compounds in the electrolyte solutions can be reduced by binding them to a graphene or h-BN nanosheet, making the organic compound–graphene or organic compound–h-BN composite a promising electrode material for lithium-ion batteries.

Received 7th January 2014
Accepted 11th March 2014

DOI: 10.1039/c4ta00103f

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1. Introduction

Lithium-ion batteries are widely used energy storage devices for portable electronics and electric vehicles due to their high-energy density and long life.^{1,2} Recently, they have been reported to be applied to grid storage.³ Most consumer batteries still use the original lithium transition-metal oxide cathodes and graphite anodes.⁴ However, toxicity, safety and resource availability are known problems with lithium transition metal oxides.¹ Alternatively, organic compounds offer new possibilities for high energy density, environmentally friendly and cheap electrode materials.^{5–8} Some of these organic molecules can be generated from biomass,⁶ preventing the depletion of limited resources. The use of organic polymers for cathode-active materials has already achieved high voltage and cycle performance comparable to those of traditional lithium-ion batteries.^{7,9,10} 9,10-anthraquinone (AQ) and its derivatives are the most energy-rich anodic materials known, but their significant solubility in electrolyte solutions influences their charge–discharge cycle number.⁵ Compared to

the pure component, AQ functionalized carbon nanocomposites possess much higher active material utilization ratios and superior ultrafast-charge and discharge abilities. For example, an AQ–carbon fabric composite has been synthesized to obtain a high performance electrode material for supercapacitors,¹¹ and an AQ–graphene nanocomposite was developed as a highly efficient electrocatalyst for the oxygen reduction reaction in alkaline conditions.¹² In all these applications, an interaction between AQ (or its derivatives) and carbon materials, especially graphene, is required in order to obtain a comprehensive understanding of the superior electrochemical properties.

Two-dimensional graphene has rather unique electronic properties such as high migration rate, massless carriers and anomalous quantum Hall effects.¹³ The unique properties of the Dirac fermions suggest the potential of graphene to be a material for improving the performance organic electrodes in lithium-ion batteries. Recently, a structure analog of graphene, *i.e.*, a sheet of heterogeneous boron nitride (BN), was successfully isolated by Pacile *et al.*¹⁴ and Han *et al.*¹⁵ Although the hexagonal BN (h-BN) sheet has the same honeycomb lattice structure as graphene, its electronic properties are significantly different from those of graphene. Monolayer h-BN is a wide band gap semiconductor. The B–N bond in the layer exhibits dipole bonding interactions, underlying the charge transfer between the B and N atoms. It is therefore interesting to carry out theoretical research and compare the

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† Electronic supplementary information (ESI) available: Additional figures. See DOI: 10.1039/c4ta00103f

binding mechanism and the binding energy of AQ and its derivatives on monolayer h-BN and graphene.

Among the various theoretical approaches, *ab initio* density functional theory (DFT) has proven to be a powerful tool for predictions of binding energy.^{16–19} Although DFT can easily deal with systems containing hundreds of atoms, the most widely used exchange-correlation DFTs, *i.e.*, local density approximation (LDA), generalized gradient approximation (GGA) and standard hybrid DFT fail to include the van der Waals (vdW) dispersion energy. However, a recent experimental and theoretical quantification of the adsorption enthalpies of organic molecules on graphene underlines the importance of non-local dispersion correlations.²⁰ The systems studied in this work contain aromatic interactions which are too large to be handled with standard wave functional theory. To accurately predict the inter- and intra-molecular non-covalent interactions, Grimme²¹ proposed a general computational scheme for the parameters used in damped atom-pairwise dispersion corrections of the form $C_6 \cdot R^{-6}$. The obtained parameters cover elements up to xenon and several common DFTs. The excellent results that have been obtained at affordable computational cost suggest vdW dispersion-corrected DFT could be a routine tool for many applications in organic chemistry or biochemistry.^{22–24} Random phase approximation²⁵ can also combat the shortcoming of LDA, GGA and hybrid DFT, but it is computationally expensive.

In this paper, a theoretical study on the adsorption of AQ and its derivatives, *i.e.*, benzofuro[5,6-*b*]furan-4,8-dione (BFFD), benzo[1,2-*b*;4,5-*b'*]dithiophene-4,8-dione (BDTD) and pyrido[3,4-*g*]isoquinoline-5,10-dione (PID), on an h-BN sheet and graphene was carried out. The vdW dispersion-corrected DFT was used to determine the binding energies for the adsorbed AQ and its derivatives. From the analysis of the calculated results, it was found that the vdW dispersion stabilization is dominant as it contributes to more than 80% of the total attraction interaction for all the systems studied. Although the binding energy for a given molecule is very close on the h-BN sheet and graphene, the preferred adsorption site is definitely different due to the great difference in atomic polarizability between the h-BN sheet and graphene. A work function shift relative to an isolated h-BN sheet or graphene was found, indicating that an interfacial dipole between the adsorbate molecule and substrate has been induced by strong physisorption.

2. Computational details

The electronic structures and energies were obtained using spin-polarized computations based on *ab initio* DFT. The GGA in the Perdew–Burke–Ernzerhof parametrization (PBE)²⁶ implemented in the DMol³ package^{27,28} was used to describe the electronic exchange and correlation effects. All core electrons were explicitly considered and some relativistic effects were introduced into the core. Version 4.4 of the double numerical plus polarization set was selected and confined within a global cutoff of 5.4 Å. Brillouin zone sampling was applied to the $4 \times 4 \times 1$ Monkhorst-pack *k*-point meshes. A 7×7 hexagonal graphene supercell with a length of 60 Å in the *z*-direction was used in all calculations. A thermal smearing with the value of 0.14 eV was applied to the orbital occupation to improve the self-

consistent field convergence. All geometries were optimized until the maximum force acting on each atom and the maximum energy change were below 0.054 eV nm^{-1} and $5.0 \times 10^{-5} \text{ eV per atom}$, respectively. The electronic density of state (DOS) was computed for the most stable configurations on a $12 \times 12 \times 1$ Monkhorst-pack *k*-point mesh. The geometric optimization process was the same as in previous work.¹⁷

Considering the robustness and computational speed of the method for long-range electron correlations, the damped atom-pairwise dispersion corrections of the form $C_6 \cdot R^{-6}$ seem most promising. They have proved to be highly accurate in many different applications.^{22–24} A general form of the vdW force contribution is²¹

$$E_{\text{vdW}} = -s_6 \sum_{i=1}^{N-1} \sum_{j=i+1}^N C_6^{ij} R_{ij}^{-6} f_{\text{damp}}(R_{ij}, R_i^0, R_j^0) \quad (1)$$

where N is the number of atoms in the system, C_6^{ij} is the dispersion coefficient for atom pair ij , s_6 is a global scaling factor that only depends on the DFT used, R_{ij} is the distance between atoms i and j , R_i^0 is the vdW radii, and $f_{\text{damp}}(R_{ij}, R_i^0, R_j^0)$ is a damping function to avoid near-singularities at small distances. In this work the C_6 coefficients and vdW radii determined by Grimme (DFT-D)²¹ are used to evaluate binding energies.

The binding energy (E_b) was calculated from

$$E_b = E_{\text{sheet}} + E_M - E_{M+\text{sheet}} \quad (2)$$

where E_{sheet} , E_M and $E_{M+\text{sheet}}$ are the energies of the pristine graphene or h-BN nanosheet, an isolated adsorbate molecule and an adsorbate molecule adsorbed on the nanosheet, respectively. The energy of an isolated adsorbate molecule was determined in a $60 \text{ Å} \times 60 \text{ Å} \times 60 \text{ Å}$ cell using a Monkhorst-Pack sampling of $5 \times 5 \times 5$.

In addition to the binding energy, an absolute potential of the Fermi level, namely, the work function, is another important quantity in understanding field emission properties. The work function is defined as the energy needed to move an electron from the Fermi level to a vacuum. In this work, it is evaluated as

$$WF = \phi_0 e - E_F \quad (3)$$

where ϕ_0 is the electrostatic potential in a vacuum away from the surface, e is the electron charge and E_F is the Fermi energy.

3. Results and discussion

Before beginning adsorption calculations, we optimized the monolayer graphene and h-BN using the PBE, and PBE-D schemes. The calculated lattice constants and work functions for monolayer graphene and h-BN are listed in Table 1. For comparison, some previous theoretical² and experimental^{29–33} results are also included in Table 1. The present work is in good agreement with the previous results. Thus the PBE-D scheme was selected for the adsorption calculations.

Adsorptions of four battery active organic molecules on all high symmetry sites of graphene and an h-BN sheet were

Table 1 Optimized lattice constant and work function of the graphene and h-BN nanosheets

	PBE	PBE-D	Expt
Lattice constant (Å)			
Graphene	2.467	2.467	2.462 ^a
h-BN	2.517	2.517	2.504 ^b
Work function (eV)			
Graphene	4.49(4.44 ^c)	4.49	4.57 ^d , 4.2 ^e
h-BN	3.67	3.65	

^a Experimental value from Kelly²⁹ and Chung.³⁰ ^b Experimental value from Solozhenko *et al.*³¹ ^c Theoretical value taken from Yu.²

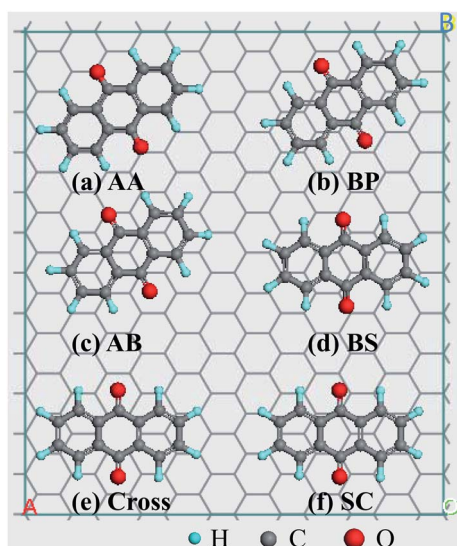
^d Experimental value from Yu *et al.*³² ^e Experimental value from Kwon *et al.*³³

schematically investigated in this work, and they are drawn in Fig. 1. The first kind of adsorption site is labeled as “AA-stacking (AA)” in Fig. 1(a), referring to all carbon atoms in the middle C₆ ring of an adsorbate molecule sitting on top of the atoms in the graphene or h-BN sheet. The second kind of adsorption site, shown in Fig. 1(b), is “bridge-parallel (BP)”, representing one bond in a graphene or h-BN sheet being under the center and parallel to the two bonds in the middle C₆ ring of the organic molecule. The third and forth kinds of adsorption site are labeled as “AB-stacking (AB)” and “bridge-slant (BS)” in Fig. 1(c) and (d), respectively. The site labeled as “Cross” represents all carbon atoms in the middle C₆ ring being located at the top of the middle of the C–C or B–N bond in a graphene or h-BN sheet. The last kind of adsorption site is a “shifted Cross (SC)” configuration, which is obtained by shifting the organic molecule in the “Cross” configuration by a half length of the C–C or B–N bond.

The calculated binding energies E_b for the four organic molecules adsorbed on monolayer graphene using the PBE and PBE-D schemes are listed in Table 2. From Table 2 one can find

that among the six adsorption sites, the binding energy of AQ and its derivatives on the BP site is the largest while the binding energy on the AA site is the smallest. The binding energies for AQ and its derivatives on graphene are in the order: BDTD $\approx$ BFFD < AQ < PID using the PBE scheme and BFFD < BDTD < AQ < PID using the PBE-D scheme. The most stable adsorption configurations for the four organic molecules on monolayer graphene are illustrated in Fig. 2. The equilibrium geometries of the four organic molecules on the different sites of monolayer graphene can be found in Fig. S1–S4.† Both the PBE and PBE-D schemes predict the same most stable configuration for AQ and its derivatives on graphene. The preferred adsorption site shown in Fig. 2 is different from that for the benzene–graphene system obtained by Chang *et al.*³⁴ They reported that the most stable configuration is the AB configuration for a benzene molecule adsorbed on monolayer graphene. This difference is not caused by the vdW interaction because of the lack of the dispersion correction in the PBE scheme. Perhaps it is due to the strong electronic dipole between the polarized bonds (C=O) in the four organic molecules inducing a charge distribution among the π electrons of carbon in graphene. It may also result from higher terms of electrostatic interaction.

The vertical distances between the organic molecules and the graphene layer can be found in Table 3. The calculated values of the vertical distance are in the range of 3.70–3.85 and 3.15–3.24 Å using the PBE and PBE-D schemes, respectively. It was found that the vertical distances are almost the same on different sites for any systems using the same scheme, and they deviate from that on the preferred site by only ± 0.01 Å. The vertical distance calculated from PBE-D is shorter than that from the PBE scheme by 0.5–0.6 Å. From Tables 2 and 3 one can conclude that the PBE functional predicts lower binding energies and relatively larger vertical distances because the π – π

**Fig. 1** Adsorption sites of an AQ molecule on a graphene or h-BN monolayer.**Table 2** Calculated adsorption energy (in eV) for the AQ, BFFD, BDTD, and PID molecules on different adsorption sites of graphene and h-BN nanosheets

Adsorbate	DFT	AA	BP	AB	BS	Cross	SC
Graphene							
AQ	PBE	0.243	0.248	0.244	0.243	0.247	0.243
	PBE-D	1.132	1.295	1.272	1.226	1.263	1.245
BFFD	PBE	0.197	0.211	0.209	0.205	0.201	0.202
	PBE-D	0.934	1.056	1.004	0.995	1.024	1.020
BDTD	PBE	0.204	0.210	0.210	0.216	0.215	0.210
	PBE-D	0.980	1.114	1.093	1.093	1.080	1.081
PID	PBE	0.249	0.258	0.256	0.250	0.254	0.250
	PBE-D	1.126	1.298	1.276	1.209	1.259	1.239
h-BN							
AQ	PBE	0.196	0.196	0.201	0.198	0.197	0.189
	PBE-D	1.154	1.262	1.305	1.257	1.297	1.309
BFFD	PBE	0.152	0.148	0.157	0.150	0.155	0.159
	PBE-D	0.956	1.040	1.057	1.029	1.028	1.100
BDTD	PBE	0.163	0.156	0.168	0.161	0.165	0.172
	PBE-D	0.996	1.088	1.108	1.074	1.075	1.146
PID	PBE	0.193	0.182	0.197	0.190	0.183	0.187
	PBE-D	1.133	1.267	1.286	1.224	1.252	1.252

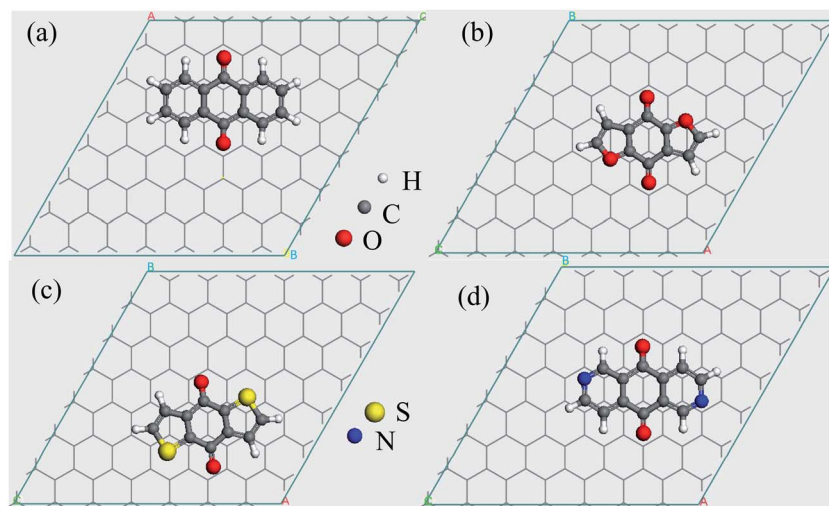


Fig. 2 Optimized structures of (a) AQ, (b) BFFD, (c) BDTD and (d) PID molecules on a graphene monolayer.

Table 3 Calculated work function shift (in eV) and vertical distance (in Å) between an adsorbed molecule and a graphene or h-BN nanosheet at the preferred adsorption site

Adsorbent	Adsorption sites	Work function shift (eV)		Vertical distance (Å)	
		PBE	PBE-D	PBE	PBE-D
Graphene					
AQ	BP	0.03	0.05	3.79	3.19
BFFD	BP	0.11	0.11	3.70	3.16
BDTD	BP	0.08	0.11	3.85	3.24
PID	BP	0.19	0.19	3.72	3.15
h-BN					
AQ	SC	1.22	1.31	4.15	3.13
BFFD	SC	1.39	1.47	3.94	3.08
BDTD	SC	1.36	1.41	4.12	3.17
PID	AB	1.52	1.55	4.11	3.08

stacking interactions between the organic molecules and graphene are not correctly described when it is used.

Fig. 3 depicts the most stable configurations of the four organic molecules adsorbed on the h-BN nanosheet using the PBE-D scheme. From the figure one can see that the SC site is the preferred adsorption site for the AQ, BFFD and BDTD molecules, while the PID molecule prefers the AB site on the h-BN nanosheet. From Table 2 one can find that the most stable adsorption configuration for the AQ-h-BN system obtained using the PBE-D scheme is different from that using the PBE scheme, indicating that the vdW force in the SC site is stronger than that on the AB site for the AQ-h-BN system. The preferred adsorption configurations of the four organic molecules on the h-BN nanosheet are definitely different from those on the graphene nanosheet. This is mainly attributed to more electrostatic attraction between the organic molecules and B atoms, and less attraction between the organic molecules and N atoms due to the polar nature of the B-N bond. In addition, the vdW

dispersion force contributes much more to the preferred adsorption configurations, especially for the AQ-h-BN system. In Table 2, the binding energies of the four organic molecules on the h-BN nanosheet are in the order: BFFD < BDTD < PID < AQ using both the PBE and PBE-D schemes. As can be seen from Table 3, the values of the vertical distances between the organic molecules and the h-BN sheet predicted using the PBE and PBE-D schemes are in the range of 3.94–4.15 and 3.08–3.17 Å, respectively. The values from the PBE-D scheme are shorter than the corresponding ones from the PBE functional for both the adsorbate-graphene and adsorbate-h-BN systems. It can be seen that the PBE functional predicts a slightly smaller binding energy on the h-BN nanosheet when compared with that on graphene. However, the PBE-D scheme predicts a very similar binding energy for each organic molecule adsorbed on monolayer graphene and h-BN, due to the π - π stacking interactions on the h-BN nanosheet being very close to those on graphene.

In order to understand the influence of the vdW dispersion on the binding mechanics the total binding energy $E_{b,PBE-G}$ is composed as

$$E_{b,PBE-G} = E_{b,PBE} + E_{b,vdW} \quad (4)$$

where $E_{b,PBE}$ and $E_{b,PBE-G}$ are the binding energies calculated using the PBE and PBE-D schemes, respectively, and $E_{b,vdW}$ is the contribution of the vdW dispersion to the binding energy. Fig. 4(a) and (b) depict the $E_{b,vdW}$ of the four organic molecules on monolayer graphene and h-BN, respectively. For all preferred adsorption sites on monolayer graphene and h-BN, the contribution of the vdW dispersion to the binding energy approximates to 1 eV. This indicates that the interactions between the four organic molecules and the graphene or h-BN nanosheet are strong physisorption. As shown in Fig. 4(a), the binding energies of the AQ, BFFD, BDTD and PID molecules on graphene predicted using the PBE-D scheme are larger than those using the PBE scheme by 1.047, 0.845, 0.904 and 1.040 eV, respectively. The values of $E_{b,vdW}$ of the four organic molecules on the

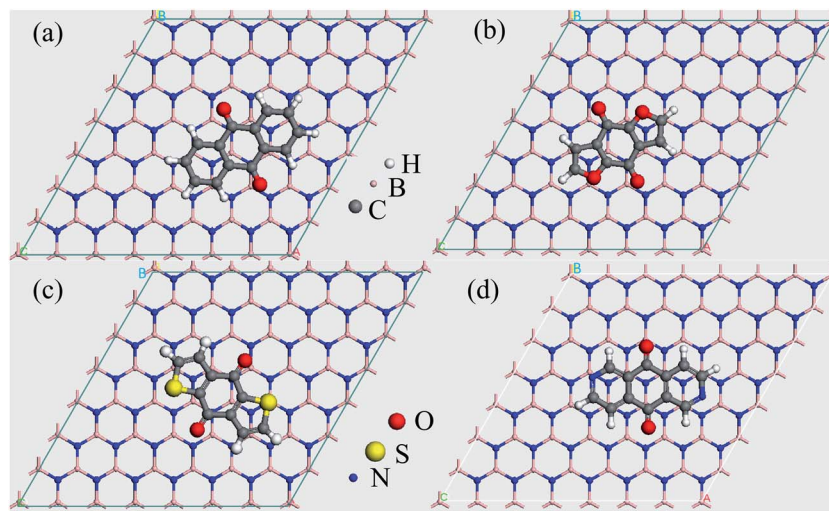


Fig. 3 Optimized structures of (a) AQ, (b) BFFD, (c) BDTD and (d) PID molecules on a h-BN monolayer.

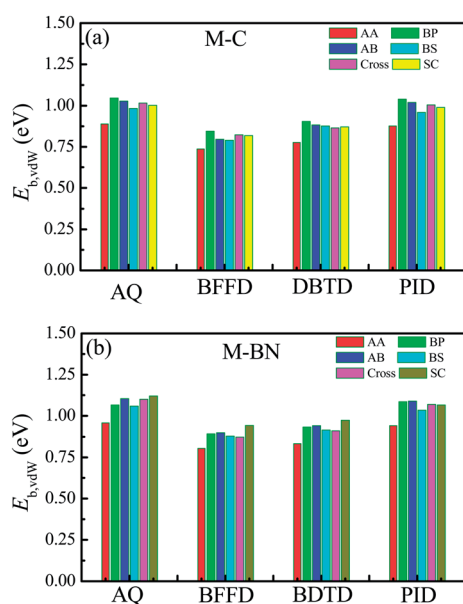


Fig. 4 Calculated vdW contributions $E_{b,vdW}$ for AQ, BFFD, BDTD and PID molecules adsorbed on graphene and h-BN monolayers.

BP site of graphene are the largest and those on the AA site are the smallest among the six adsorption sites. Fig. 4(b) shows that the value of $E_{b,vdW}$ on the SC site of the h-BN sheet is the largest for the AQ-h-BN, BFFD-h-BN, and BDTD-h-BN systems, while that on the AB site is the largest among the six sites for the PID-h-BN system. Similar to the systems of graphene, the contribution of the vdW dispersion is the smallest when the organic molecules are adsorbed on the AA site. The PBE-D scheme predicts $E_{b,vdW} = 1.108, 0.941, 0.974$ and 1.089 eV for the AQ-h-BN, BFFD-h-BN, BDTD-h-BN and PID-h-BN systems, respectively. Overall, the largest value of $E_{b,vdW}$ corresponds to the preferred adsorption site on monolayer graphene and h-BN. The magnitudes of $E_{b,vdW}$ for each organic molecule on the

graphene and h-BN nanosheets are very close to each other, indicating that the value of $(C_6^{M-B} + C_6^{M-N})/2$ is quite near to the value of C_6^{M-C} , where M-B, M-N and M-C refer the atomic pairs between the constituent atoms in the organic molecule and the B, N and C atoms, respectively. If the dispersion contribution (in %) is calculated as $E_{b,vdW}/E_{b,PBE-G}$, it contributes to more than 80% of the total attractive interaction for all the complexes considered in this work.

The binding energies are related to the battery life when organic compounds are used in an electrode. The strong binding energies between AQ (and its derivatives) and graphene or h-BN can reduce the chemical potential of AQ (and its derivatives). According to thermodynamic principles, a compound with a lower chemical potential has a smaller solubility in a specific solvent. Therefore, introduction of graphene or h-BN to AQ (and its derivatives) can prevent unwanted dissolution in the non-aqueous electrolyte of lithium-ion batteries that is suffered by AQ and its derivatives, and thereby increase the charge-discharge cycle number of the lithium-ion batteries. This makes organic compound-graphene or organic compound-h-BN composites promising electrode materials for lithium-ion batteries. The above theoretical conclusion is expected to be validated by related experimental data. Although there is no experimental data that can be used directly to validate this theoretical result, a recent experimental result reported by Song *et al.*³⁵ showed that nanocomposites combining graphene with two promising polymer cathode materials, poly(antraquinonyl sulfide) and polyimide, achieved better cycling stability.

The total electronic density of state (DOS) and projected DOS (PDOS) for the four organic molecules adsorbed on the graphene and h-BN nanosheets are plotted in Fig. 5 and 6, respectively. The basic shape of the DOS remains almost the same with respect to the pristine sheet. There is no evidence for hybridization between the four organic molecules and the intrinsic graphene or h-BN nanosheet. From analysis of the Mulliken charges using the PBE-D scheme, one can find a very small charge transfer between any of the four organic molecules

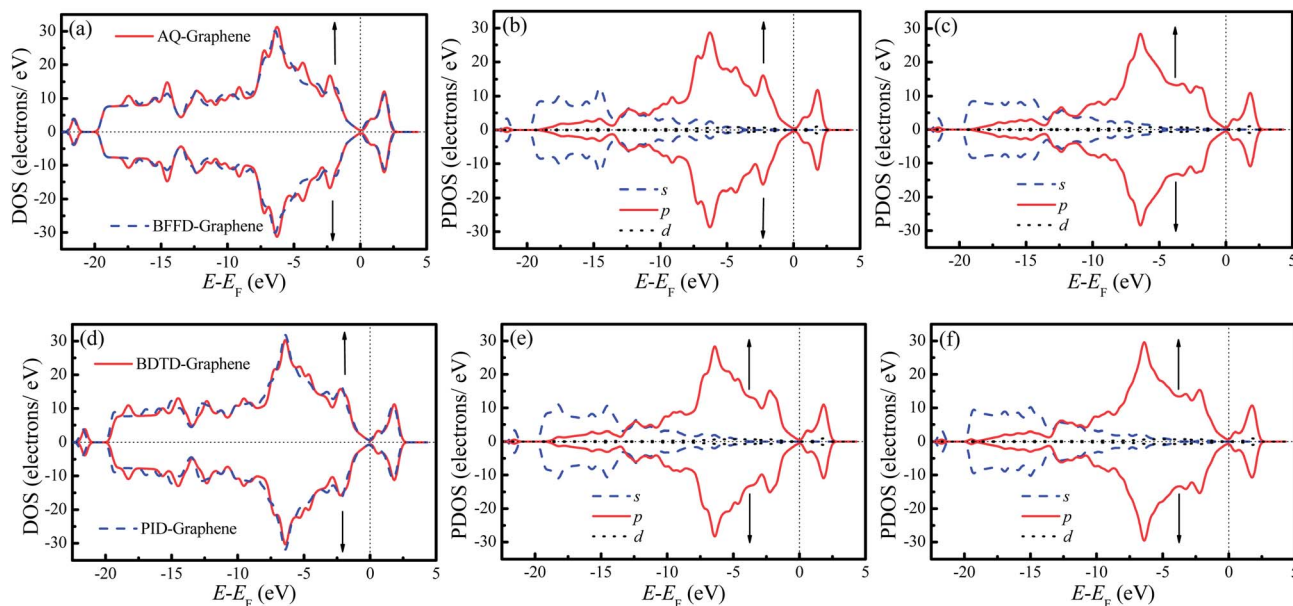


Fig. 5 Total and projected DOS for the most stable adsorption configurations from the PBE-D scheme: (a) DOS for the AQ-graphene and BFFD-graphene systems, (b) PDOS for the AQ-graphene system, (c) PDOS for the BFFD-graphene system, (d) DOS for the BDTD-graphene and PID-graphene systems, (e) PDOS for the BDTD-graphene system and (f) PDOS for the PID-graphene system. The arrows denote spin-up (↑) and spin-down (↓) states.

and the graphene or h-BN nanosheet. The distribution of charge difference in real-space $\Delta\rho(\mathbf{r})$ can be obtained from

$$\Delta\rho(\mathbf{r}) = \rho_{\text{M+sheet}}(\mathbf{r}) - \rho_{\text{M}}(\mathbf{r}) - \rho_{\text{sheet}}(\mathbf{r}) \quad (5)$$

where $\rho_{\text{M+sheet}}(\mathbf{r})$, $\rho_{\text{M}}(\mathbf{r})$ and $\rho_{\text{sheet}}(\mathbf{r})$ are the charge density distributions in real-space of the molecule-substrate system, organic

molecule and isolated graphene or h-BN nanosheet, respectively. Fig. 7(a) and (b) depict $\Delta\rho(z)$ calculated from the PBE-D scheme for an AQ molecule adsorbed on the graphene and h-BN nanosheets, respectively. Because the electrostatic interactions are important for the work function, the planar averaged electrostatic potentials calculated from the PBE-D scheme for the AQ-graphene and AQ-h-BN systems are plotted in Fig. 8(a) and (b),

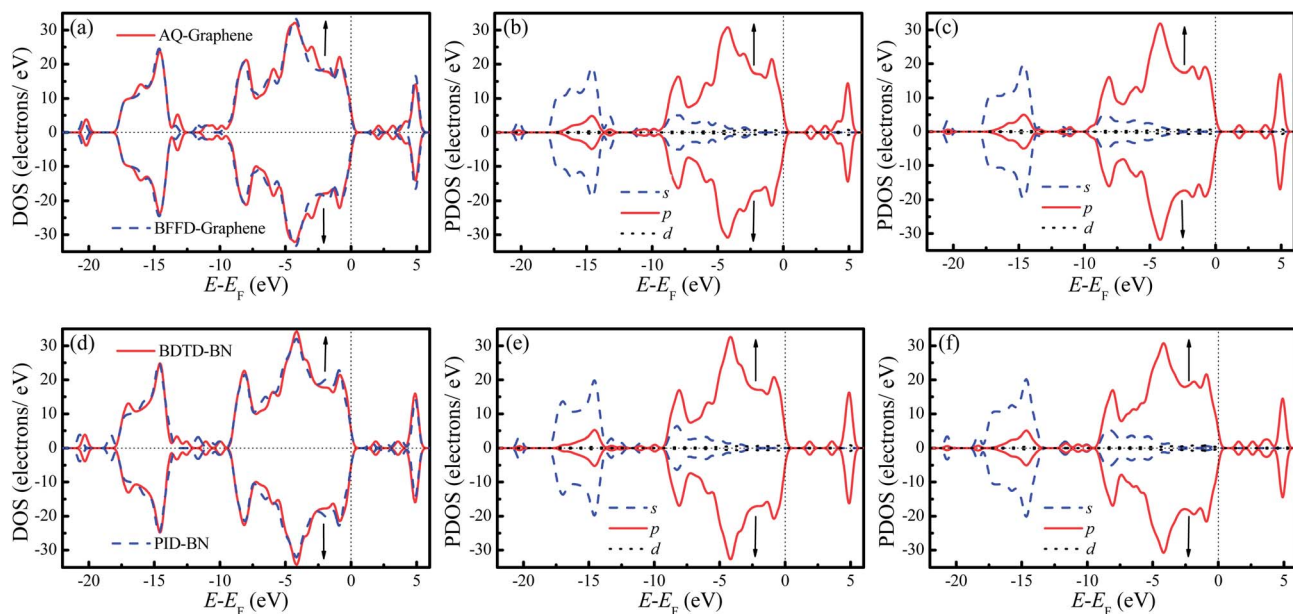


Fig. 6 Total and projected DOS for the most stable adsorption configurations from the PBE-D scheme: (a) DOS for the AQ-h-BN and BFFD-h-BN system, (b) PDOS for the AQ-h-BN system, (c) PDOS for the BFFD-h-BN system, (d) DOS for the BDTD-h-BN and PID-h-BN systems, (e) PDOS for the BDTD-h-BN system and (f) PDOS for the PID-h-BN system. The arrows have the same meaning as in Fig. 5.

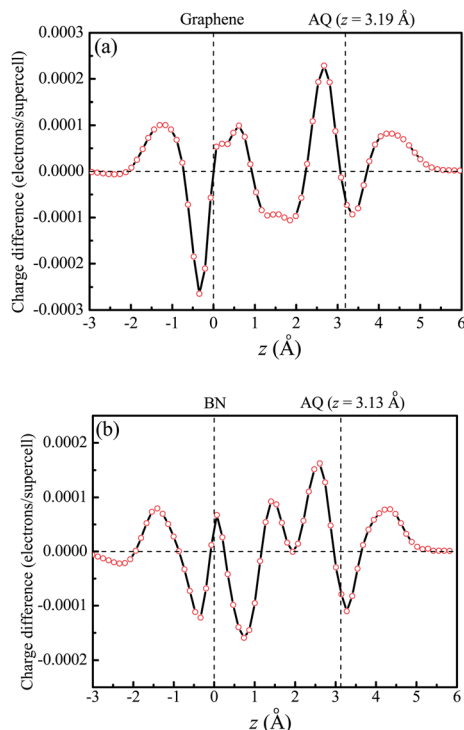


Fig. 7 Charge difference profiles between the adsorption system and the isolated systems along the z -direction for (a) the AQ-graphene and (b) the AQ-h-BN systems. The vertical dashed lines represent the position of the monolayer ($z = 0$) and AQ molecule.

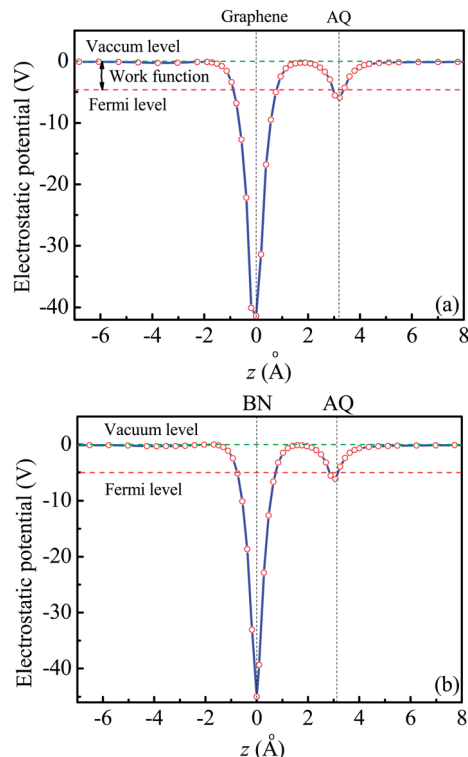


Fig. 8 The planar averaged electrostatic potentials along the z -direction for (a) the AQ-graphene and (b) the AQ-h-BN systems. The vertical dashed lines have the same meaning as in Fig. 7.

respectively. An interfacial dipole can result from charge rearrangement at the interface. This dipole causes a portion of the work function to shift. All work function shifts for the complexes studied in this work can be found in Table 3. As can be seen from the table, the work function shift is almost the same using the PBE and PBE-D schemes for each organic molecule on graphene. In contrast, for the h-BN monolayer, the work function shift from the PBE functional is smaller than that from the PBE-D scheme. The work function shifts obtained are in the order: AQ < BDTD < BFFD < PID for the organic molecules adsorbed on both graphene and h-BN nanosheets. The work function shift ΔWF can be understood from the expression given by

$$\Delta WF = \Delta\mu - ep/(A\epsilon_0) \quad (6)$$

which is a modification of the work function expression proposed by Wigner and Bardeen,³⁶ and later by Lang and Kohm.³⁷ In eqn (6), $\Delta\mu$ is the difference of the bulk chemical potential of electrons between the adsorption complexes and the pristine graphene or h-BN sheet, and the second term on the right hand side is the energy necessary to penetrate the dipole barrier at the surface, where p is the interfacial dipole, A is the area of the surface and ϵ_0 is the permittivity of free space. For the systems studied in this work, $\Delta\mu$ is strongly dependent on the work function of the isolated adsorbate parallel to the xy -plane of the supercell. The work functions for the AQ, BFFD, BDTD and PID molecules obtained using the PBE-D scheme are 5.01, 5.28, 5.21 and 5.41 eV, respectively. These values dominate

the sequence of the work function shift for the systems involved in this work. This conclusion is consistent with the results from the surface coverage work function model proposed by Longo *et al.*³⁸ Of course, the surface dipole also contributes to the work function shift *via* the electrostatic relation included in eqn (6).

The treatment of spin-unrestricted systems may be improved further since the results obtained from the present PBE tend to favor delocalized solutions. The negative ionization potentials (IPs) and electron affinities of a test set of molecules were used by Zhang and Musgrave³⁹ to estimate the accuracies of 11 DFT functionals including the local spin density approximation (LDA), five GGAs, three hybrid GGA functionals, one hybrid functional, and one hybrid meta GGA functional. They found that the highest occupied molecular orbital (HOMO) eigenvalues from KMLYP,⁴⁰ BH&HLYP,⁴¹ B3LYP,⁴² PW91,⁴³ PBE²⁶ and BLYP⁴⁴ predict negative IPs with average absolute errors of 0.73, 1.48, 3.10, 4.27, 4.33 and 4.41 eV, respectively. The GGAs predict the HOMO-lowest unoccupied molecular orbital (LUMO) gap relatively accurately with an average absolute error of approximately 0.73 eV. Therefore, other DFT methods such as hybrids (KMLYP,⁴⁰ BH&HLYP,⁴¹ or B3LYP⁴²) may therefore improve the calculated binding energies as well as the binding energy orders.

4. Conclusions

The adsorption of the AQ, BFFD, BDTD and PID molecules on monolayer h-BN and graphene has been investigated using both

the PBE and PBE-D schemes. The binding energies obtained using the PBE-G scheme were larger than those using the PBE scheme. The contribution of the vdW dispersion to the binding energy ranges from 0.95 to 1.11 eV, and contributes to more than 80% of the total attractive interaction for all the complexes studied in this work. In addition, they are close to each other for the four organic molecules on the h-BN and graphene nanosheets. The calculated binding energies are in the order: BFFD < BDTD < AQ < PID on the graphene nanosheet and BFFD < BDTD < PID < AQ on the h-BN nanosheet. The inclusion of the vdW dispersion correction results in a slightly larger work function shift and a shorter vertical distance between the organic molecules and the graphene or h-BN nanosheet when compared with those using the PBE functional. The strong physisorption induces an interfacial dipole between the organic molecules and the graphene or h-BN nanosheet, resulting in a portion of work function shift relative to the isolated graphene or h-BN nanosheet. The work functions of the four organic molecules dominate the sequence of the work function shift as AQ < BDTD < BFFD < PID. The calculated results indicate that AQ and its derivatives, *i.e.*, BFFD, BDTD and PID, can be strongly bound to a graphene or h-BN nanosheet to reduce their solubility in electrolyte solutions and enlarge the cycle number of batteries when they are used for electrode-active materials. Therefore, addition of graphene or h-BN nanosheets to materials composed of the AQ, BFFD, BDTD or PID molecules will improve their performance as environmentally friendly organic electrode materials for lithium-ion batteries.

Acknowledgements

Here I would like to make an acknowledgement to Yong-Jun Du for her effort in collecting calculated data and preparing the manuscript. The author also greatly appreciates the financial support of the National Natural Science Foundation of China (no. 21376131).

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